L. D. COLLEGE OF ENGINEERING

(Affiliated to Gujarat Technological University)

Navrangpura, Ahmedabad-380015

A Project Report

On

"GreenPulse - Sustainable Energy Transition Platform"

for

Sustainable Energy Technology (BE05000561)

B.E. III, SEMESTER – V

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Academic Year: 2026–27

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This is to certify that the project entitled "**GreenPulse - Sustainable Energy Transition Platform**" has been carried out by **Pethani Aesa Rajeshbhai (240280107114)**, **Prajapati Yash Mahendrabhai (240280107119)**, **Vaja Vinay Jentibhai (240280107147)** and **Viradiya Kush Prakashbhai (240280107154)** under my guidance in partial fulfillment of the course **Sustainable Energy Technology (BE05000561)** in 5th Semester of Bachelor of Computer Engineering during the academic year 2026-27.

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CANDIDATE'S DECLARATION

We have finished our project report entitled "**GreenPulse - Sustainable Energy Transition Platform**" and submitted it to our respective guide. We are in the 5th semester and we have tried to give our best. We have done our work honestly and in a good way.

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ABSTRACT

The world needs more energy every year, but burning coal and gas harms our climate. To solve this, we must use more clean energy such as sun, wind and water. This report presents **GreenPulse**, a website that helps students understand how clean energy systems work. It was made as a Project-Based Learning project for the subject Sustainable Energy Technology (BE05000561) at L. D. College of Engineering, Ahmedabad.

With GreenPulse, a user can build a small clean-energy power system (a "microgrid") using solar panels, wind turbines, a small hydro plant, a biomass plant and a battery. The user moves sliders and instantly sees how much electricity is produced, how much money is saved and how much pollution (CO_2) is avoided. The website is built with React, Three.js and other web tools. It needs no server or database, because all calculations run inside the user's browser using simple energy formulas.

The website has eight pages that cover all five units of the GTU syllabus: a Dashboard with a clean-energy score (SDG 7), a Simulator with a 3D model of the power system, a page on green hydrogen and fuel cells, a waste-to-energy calculator, an energy audit tool, a technology comparison page, a talking AI assistant, and an About page. The project is live on the internet and shows that difficult energy topics can be learned in an easy, visual and free way.

Keywords: sustainable energy, renewable microgrid, digital twin, SDG 7, green hydrogen, waste-to-energy, energy audit, React, Three.js

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CHAPTER 1
INTRODUCTION

1.1 Background & Motivation

Electricity is needed everywhere: in homes, schools, hospitals and factories. But most of the world's electricity still comes from burning coal, oil and gas, which release carbon dioxide (CO_2) and cause climate change. In India, one unit (1 kWh) of grid electricity releases about 0.82 kg of CO_2. So every unit made from the sun, wind or water instead of coal helps the environment.

Students and engineers need to understand how clean energy systems behave. How much electricity does a solar panel really give in a day? When do we need a battery? How much money and pollution can we save? These things are hard to understand only from books. This is why our team built GreenPulse, a website where these ideas can be seen, tried and explored with a few clicks.

1.2 Problem Statement

Most online energy tools are plain calculators or spreadsheets. They show a number but do not explain how the electricity is made or where energy is lost. Professional tools are very powerful, but they must be installed, can be costly and need special training. Students need a free tool that runs in the browser, shows things visually and explains the science in a simple way. GreenPulse is made to fill this gap.

1.3 Objectives

- To build a simulator for solar, wind, hydro and biomass energy together with a battery.
- To keep the calculation of each energy source in its own small, simple code file.
- To show the power system as a live 3D model and an animated flow diagram.
- To measure sustainability using CO_2 saved, money saved and an SDG 7 score out of 100.
- To cover hydrogen, waste-to-energy and energy audit as required by the GTU syllabus.
- To compare eight energy technologies side by side.
- To add a talking AI assistant and put the whole project online for everyone.

1.4 Scope of the Project

GreenPulse is an educational tool. It simulates one typical day of a small power system. It includes eight web pages, the calculation code, the 3D model, the voice assistant and online deployment. It does not include things like detailed weather data, full financial planning, user accounts or saving data. The formulas are simplified for learning, so the results should not be used for real engineering design.

CHAPTER 2
LITERATURE REVIEW & BACKGROUND THEORY

2.1 Renewable Energy Systems Overview

Renewable energy comes from natural sources that never run out. **Solar panels** turn sunlight directly into electricity. **Wind turbines** use moving air to spin a generator. **Hydropower** uses falling water to turn a turbine, and it is the oldest and most reliable clean source. **Biomass** means using farm and plant waste to make heat and electricity. Sun and wind are not available all the time, so a **battery (BESS)** is used to store extra electricity and give it back when needed.

2.2 UN Sustainable Development Goal 7 (SDG 7)

The United Nations has set 17 goals for a better world by 2030. SDG 7 says everyone should have "affordable and clean energy". This means electricity for all, more renewable energy and better energy saving. SDG 7 is also linked to SDG 9 (industry and innovation), SDG 11 (sustainable cities), SDG 12 (responsible consumption) and SDG 13 (climate action). GreenPulse gives each design a score out of 100 so that students can see how close it is to these goals.

2.3 Microgrid Architecture & Energy Management Systems

A microgrid is a small local power system with its own generators, battery and users. It can work together with the main grid or alone. A controller called the **Energy Management System (EMS)** decides where the electricity should go. The usual rule is: first use clean energy for the demand, store any extra in the battery, use the battery when clean energy is less, and take from the main grid only at the end. GreenPulse follows this same rule.

2.4 Digital Twin Technology in Energy Systems

A digital twin is a computer copy of a real system. It is used by companies to watch wind farms and power stations and to find problems early. In GreenPulse, the digital twin is a 3D model of a campus power system. When the user changes a slider, the 3D model and the numbers change together, so the picture always matches the calculation.

2.5 Green Hydrogen & Fuel Cell Technologies

Green hydrogen is made by splitting water into hydrogen and oxygen using clean electricity. The machine that does this is called an **electrolyzer**. There are three main types: **AWE** (alkaline, cheap and proven), **PEM** (reacts quickly to changing power) and **SOEC** (very efficient but works at very high temperature). Hydrogen can later be turned back into electricity in a **fuel cell**. Some energy is lost in both steps, so only about 30–40 % of the original electricity comes back. Because of this, hydrogen is best used for long-time storage and for industries that cannot use electricity directly.

2.6 Waste-to-Energy Technologies

Waste-to-energy means getting useful energy from garbage and farm waste, which also reduces landfills. **Thermal methods** use heat: incineration (burning), pyrolysis (heating without air) and gasification (heating with little air to make fuel gas). **Biological methods** use tiny living organisms: anaerobic digestion makes biogas from food and farm waste, and landfill gas recovery collects gas from old garbage dumps. Wet waste gives less energy, and waste with more organic matter suits biological methods better.

2.7 Energy Audit Methodology

An energy audit is a check of how a building uses electricity. First, list all the equipment. Then note the power rating and the hours used per day. Multiply them to get daily energy use, and multiply by the tariff to get the cost. Finally, find the biggest users and suggest ways to save. The

basic rule is simple: energy = power × time. This is exactly what the GreenPulse Energy Audit page does.

2.8 Related Work / Existing Simulators

Some well-known tools already exist. HOMER is used to design microgrids, PVWatts estimates solar output, and RETScreen checks if clean energy projects are worthwhile. They are accurate, but they are made for professionals and they mostly give tables and reports. GreenPulse is different because it is made for learning: it is free, needs no installation and explains things visually. Table 2.1 compares them.

Table 2.1: GreenPulse compared with other tools

Point	Professional tools	Spreadsheets	GreenPulse
Installation	Software or account needed	File download	None, runs in browser
Cost	Licence or limited access	Free	Free (MIT licence)
Explanation	Only results and reports	Formulas hidden	Step-by-step visuals
Visuals	Charts and tables	Basic charts	3D model, animated flow
Learning help	Manuals	None	SDG score, voice assistant
Made for	Design engineers	Anyone	Students and teachers

CHAPTER 3
SYSTEM DESIGN & ARCHITECTURE

3.1 Overall System Architecture

GreenPulse copies a real small power system. Four sources (solar, wind, hydro and biomass) send electricity to a central controller. The controller can charge or use the battery, supply the users, and buy from or sell to the main grid. Figure 3.1 shows this flow, and the same flow is animated on the Simulator page.

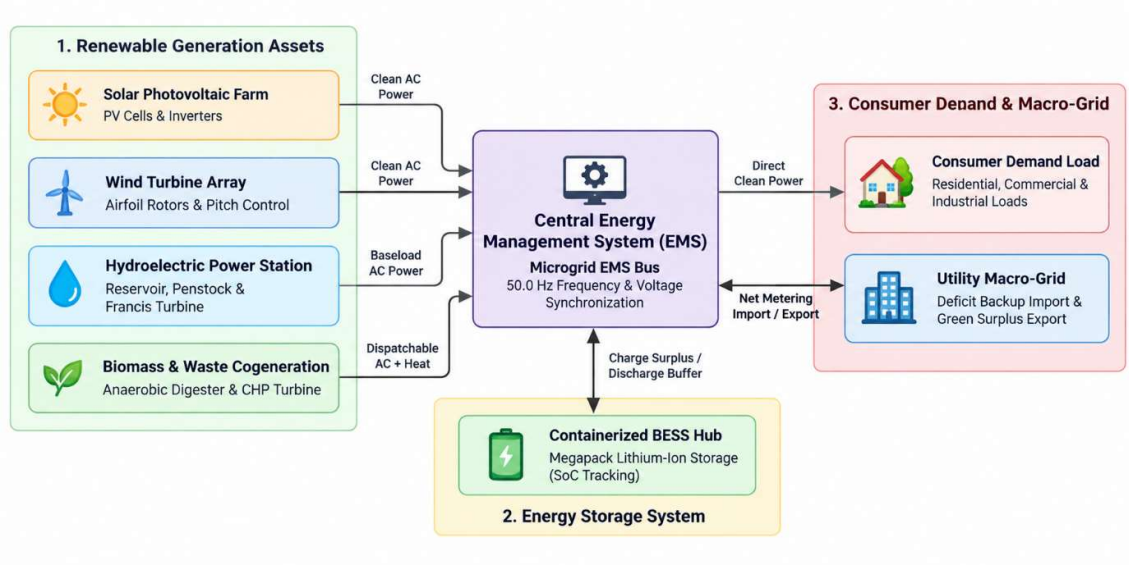


Figure 3.1: Energy-flow architecture of GreenPulse

3.2 Technology Stack Selection & Justification

A "technology stack" is the set of tools used to build the website. Table 3.1 lists them with a simple reason for each.

Table 3.1: Tools used

Area	Tool	Why it is used
Main framework	React 19	Builds the website from small reusable parts
Build tool	Vite 8	Runs the project quickly while developing
Page navigation	React Router v7	Moves between the 8 pages without reloading
3D graphics	Three.js	Draws the 3D model in the browser
Animation	Framer Motion, GSAP, Lenis	Smooth page changes, flowing energy and scrolling
Charts	Recharts	Draws area, bar, donut and radar charts
Small extras	Lucide React, react-countup	Icons and counting-up numbers
Styling	CSS, TailwindCSS	Colours, layout and the glass-like look
Code checking	OxLint	Finds mistakes in the code quickly
AI voice	Google Gemini, Web Speech API	Speaks the briefing aloud
Hosting	Vercel	Puts the website online automatically

All these tools are free and widely used, so there is plenty of help available online. The code is written in JavaScript.

3.3 Application Architecture (SPA Design)

GreenPulse is a Single-Page Application (SPA). The browser loads the website only once, and after that, clicking a menu link just changes the content on the screen without reloading. This makes the website fast and smooth. Table 3.2 lists the eight pages.

Table 3.2: Pages of the website

Address	Page	Syllabus unit
/	Dashboard	Unit 1: Energy Sustainability & SDGs
/simulator	Simulator & 3D Digital Twin	Unit 2: Renewable Energy Systems
/emerging-technologies	Emerging Technologies	Unit 3: Hydrogen & Fuel Cells
/waste-to-energy	Waste → Energy	Unit 4: Waste-to-Energy
/energy-audit	Energy Audit	Unit 5: Energy Management & Audit
/compare	Compare	Technology selection matrix
/sdg-impact	SDG Impact	SDG 7 / 9 / 11 / 12 / 13 chart
/about	About	Project details and syllabus mapping

3.4 Component Architecture

The code is split into pages, shared parts (components), shared data (context), calculation files (utils) and fixed data. Seven shared components are used on many pages.

Table 3.3: Shared components

Component	Job it does
DigitalTwin3D.jsx	Draws the 3D campus (the biggest file, about 40 KB)
EnergyFlowDiagram.jsx	Animated diagram of energy moving through the system
VoiceAssistantWidget.jsx	The Mitra and Miles voice assistant
KPICard.jsx	A card showing one key number
SDGScoreRing.jsx	Circular meter for the SDG 7 score
SliderInput.jsx	A slider to change a setting
AnimatedNumber.jsx	Makes numbers count up smoothly

3.5 State Management Strategy

"State" means the information the website remembers, such as the current slider values. GreenPulse keeps this in two shared places. **EnergyContext** stores all slider values and the results calculated from them. Whenever a slider moves, all pages update immediately. **VoiceContext** remembers which voice (Mitra or Miles) is chosen and whether it is speaking. We did not use bigger tools like Redux because the project is small and React's built-in feature is enough.

3.6 AI Voice Copilot Architecture (Mitra & Miles)

The voice assistant reads the current results and speaks a short briefing. It has two characters. **Mitra** is a friendly sustainability advisor who talks about CO_2 saved and the SDG 7 score. **Miles** is a grid operator who talks about power produced, battery level and supply to users. The main voice comes from Google's Gemini AI. If there is no internet or no API key, the website automatically switches to the browser's own built-in voice, so the feature always works.

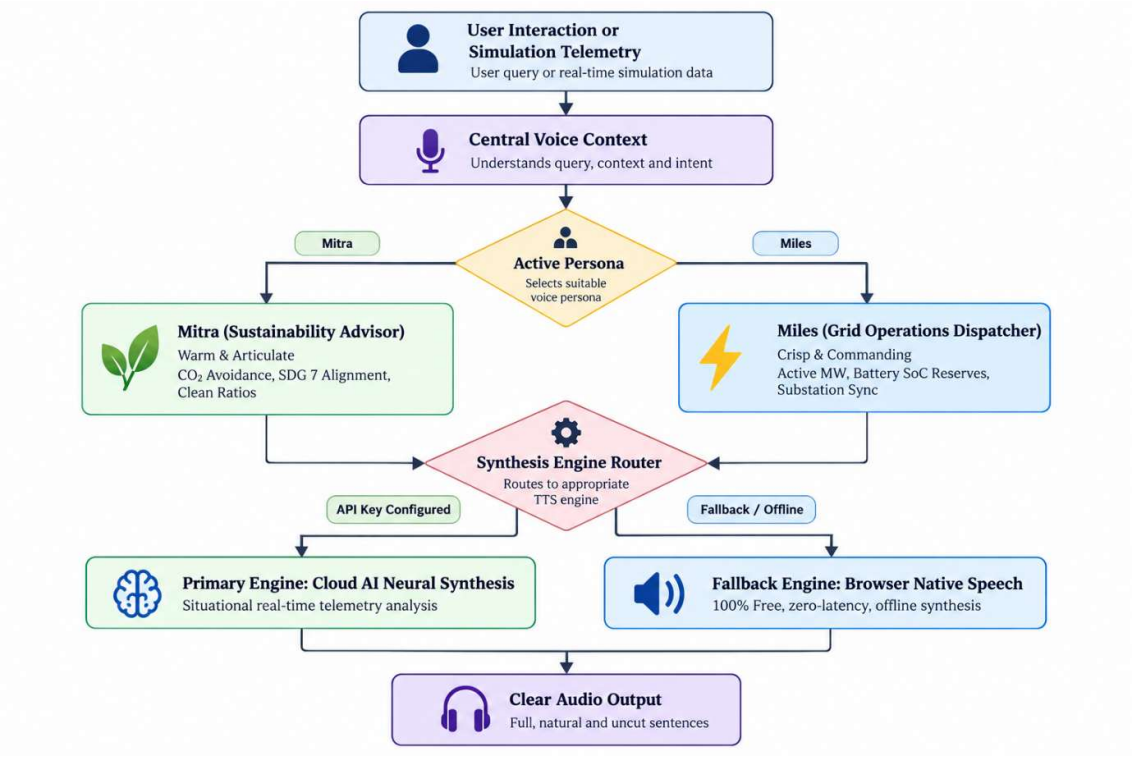


Figure 3.2: AI Voice Copilot flow

3.7 Database & Storage Design

GreenPulse has no server and no database, and this is a deliberate choice. Every answer is calculated directly from the slider values, and there is no personal data to store. This makes the website very fast, free to host and easy to maintain. The only drawback is that settings are lost when the page is refreshed, which we plan to fix in the future.

3.8 Deployment Architecture (Vercel)

The code is stored on GitHub (Vinay-vaja/SET_project). Vercel is connected to it, so whenever new code is uploaded, Vercel automatically builds and publishes the website. The live website is at https://set-visulizer-mopec-ldce.vercel.app. This means any improvement reaches users within minutes without extra work.

CHAPTER 4
IMPLEMENTATION

4.1 Development Environment Setup

The project was built using Node.js and npm, Vite for running and building, Git and GitHub for sharing code in the team, and Visual Studio Code as the editor. All the libraries (React, Three.js, GSAP, Recharts and others) are listed in the file package.json and are installed with one command. Steps are given in Appendix B.

4.2 Module 1: Dashboard (Unit 1 — SDG Alignment)

The Dashboard is the home page. It shows the main results in one view. Five cards show total energy produced (kWh/day), share of clean energy (%), CO_2 avoided (kg/day), money saved (₹/day) and dependence on the main grid (%). Other parts of the page are a donut chart of the energy mix, the SDG 7 score meter, a live panel comparing generation and demand with the battery level, a 24-hour chart and a 7-day chart.

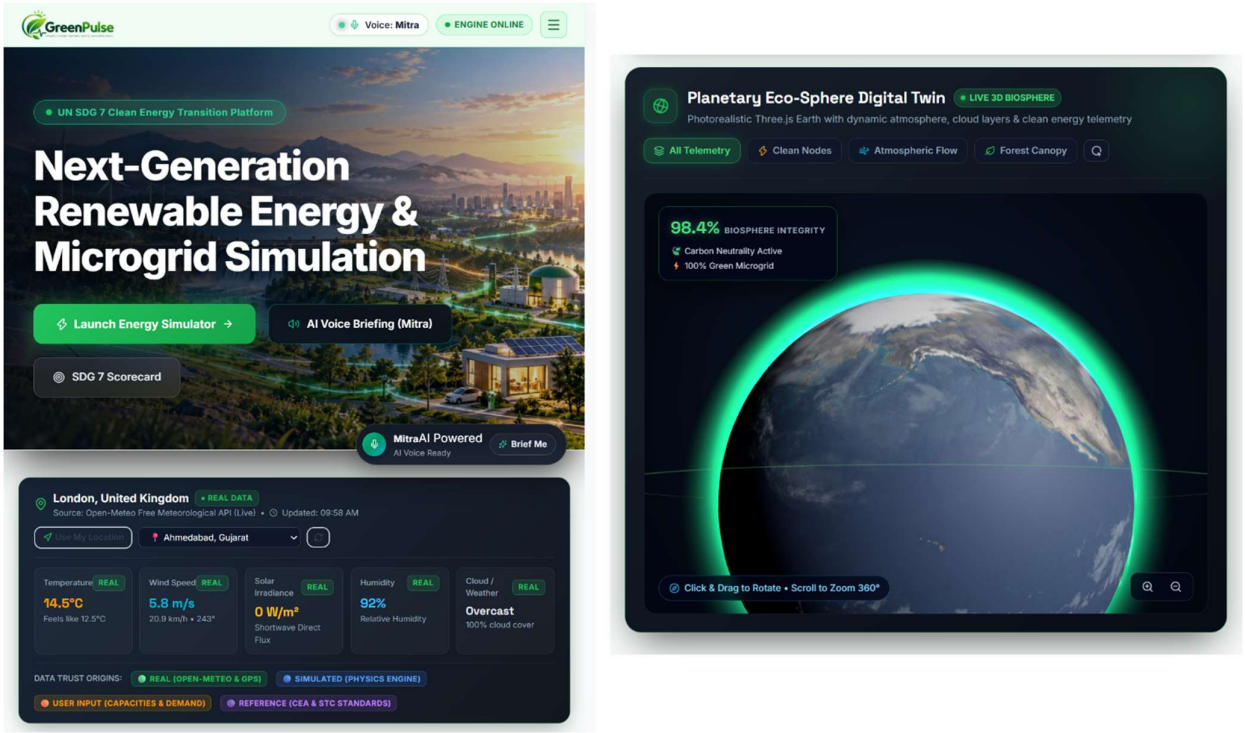


Figure 4.1: Dashboard page

4.3 Module 2: Renewable Energy Simulator (Unit 2)

The Simulator is the heart of the project. The user changes settings with sliders, and the results instantly update the 3D model, the animated diagram and the explanation panels. Table 4.1 lists all the settings with their starting values and limits.

Table 4.1: Simulator settings

Part	Setting	Default	Range
Solar PV	Installed capacity (kW)	10	0–100

Part	Setting	Default	Range
	Panel efficiency (%)	18	10–25
	Average sunlight (h/day)	5.5	1–10
Wind	Turbine capacity (kW each)	5	1–50
	Number of turbines	2	0–20
	Average wind speed (m/s)	7	1–20
	Capacity factor (%)	30	10–50
Hydro	Capacity (kW)	3	0–50
	Operating hours (h/day)	8	1–24
	Efficiency (%)	80	50–95
Biomass	Input (kg/day)	50	0–500
	Conversion efficiency (%)	25	10–50
Battery	Capacity (kWh)	20	0–200
	Starting charge (%)	50	0–100
	Charge / discharge efficiency (%)	92 / 92	70–99
Grid & load	Daily demand (kWh)	60	5–500
	Peak load (kW)	8	1–100
	Grid price (₹/kWh)	7	1–20

4.3.1 Solar PV Physics Engine

The solar file calculates the electricity from the panel size, sunlight hours and efficiency. Real panels also lose a little power when they get hot, and this is included in the full formula. In the 3D model, tilted solar panels are shown, and the explanation panel shows the steps: sunlight is captured, DC power is made, it is converted to AC, and it is delivered.

4.3.2 Wind Turbine Aerodynamic Engine

The wind file calculates daily energy from turbine size, number of turbines and capacity factor. It also explains four situations: too little wind (turbine waits), normal wind (power is produced), strong wind (full power) and storm (brakes are applied for safety). The 3D turbines rotate on screen.

4.3.3 Hydroelectric Station Engine

The hydro file uses the plant size, running hours and efficiency to find daily energy. The explanation panel shows how water falls from the reservoir through a pipe (penstock), spins the turbine and runs the generator. The 3D model shows the pipe and the power house.

4.3.4 Biomass CHP Engine

The biomass file finds the energy from the daily amount of plant waste and the conversion efficiency. Wet waste gives less energy, and this is considered. The waste is heated to about 850 °C to make fuel gas. If the waste heat is also used (CHP means combined heat and power), the total efficiency goes above 72 %.

4.3.5 Battery Storage (BESS) Module

The battery file follows a simple rule. If there is extra clean electricity, the battery is charged (a little energy is lost while charging). If there is a shortage, the battery supplies power first, and only then the grid. LED lights on the 3D battery show its charge level.

4.3.6 3D Digital Twin (Three.js)

The 3D model shows a small campus with solar panels, wind turbines, a hydro station, a biomass plant and a battery yard connected by roads. There are three lighting modes (Daylight, Sunset and Night) and five quick camera views (Overview, Solar PV, Wind Farm, BESS Hub, Hydro & Bio). Users can also rotate, zoom and move around freely, or go full screen.

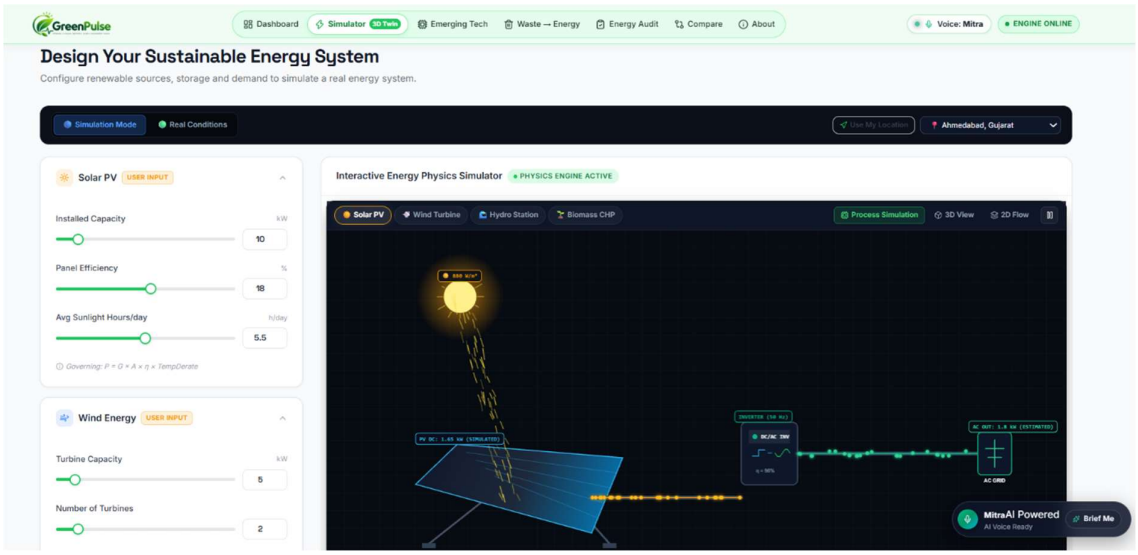
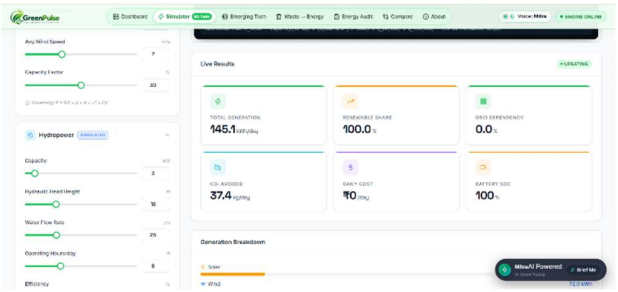
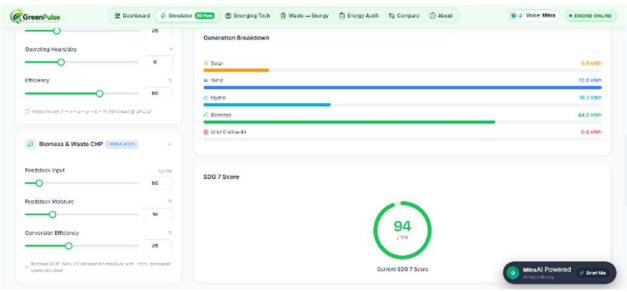


Figure 4.2: Simulator page with settings panel and 3D Digital Twin viewport





4.4 Module 3: Emerging Technologies (Unit 3 — Hydrogen)

This page explains green hydrogen as a simple chain: electricity → electrolyzer → hydrogen storage → fuel cell → output. Tabs compare electrolyzer types (AWE, PEM, SOEC) and fuel cell types (PEMFC, SOFC, AFC, PAFC). A calculator takes the electricity input and efficiencies and shows how much hydrogen is made, how much electricity comes back, the round-trip efficiency, and the CO_2 saved compared with the usual method of making hydrogen from natural gas (SMR).

Table 4.2: Electrolyzer and fuel cell types (typical values)

Technology	Working temperature	Efficiency	Short note
AWE electrolyzer	60–80 °C	60–70 %	Proven, low cost, long life
PEM electrolyzer	50–80 °C	65–75 %	Reacts fast to changing power
SOEC electrolyzer	700–850 °C	80–90 %	Very efficient; still new

Technology	Working temperature	Efficiency	Short note
PEMFC	60–80 °C	40–60 %	Vehicles and portable use
SOFC	600–1000 °C	50–65 %	Good for heat + power
AFC	60–90 °C	50–60 %	Needs very pure gases
PAFC	150–200 °C	40–45 %	Used in power units

4.5 Module 4: Waste-to-Energy Calculator (Unit 4)

This page covers five processes: incineration, pyrolysis and gasification (heat-based), and anaerobic digestion and landfill gas recovery (biological). The user enters the amount of waste, the organic share, the moisture and the efficiency. The page shows electricity produced per day, CO_2 avoided and the waste processed, along with the basic chemistry of the process.

4.6 Module 5: Energy Audit Tool (Unit 5)

This page works like a digital audit sheet. The user sets the electricity price and adds equipment such as ACs, computers, fridges, fans and lights, with quantity, power and hours of use. The page then shows daily and monthly energy use, the monthly bill, the carbon footprint and a chart of the biggest users. An AI tips section points out heavy users (for example, an AC using over 60 % of the energy) and suggests ways to save, with estimated savings in units and rupees.

4.7 Technology Comparison Matrix

The Compare page lets the user pick 2 to 5 of eight technologies: Solar, Wind, Hydro, Biomass, Coal, Nuclear, Natural Gas and Green Hydrogen. They are compared on CO_2 emissions, efficiency, availability, set-up cost, running cost, scalability and environmental impact. The page gives badges like “Best for Low Emissions”, “Best for High Efficiency” and “Best for Low Capital Cost”.

4.8 AI Voice Studio Integration

The voice assistant opens from the top menu or from the “Brief Me” buttons on each page. It collects the current results, sends them to the chosen character (Mitra or Miles) and speaks the answer. If the Gemini AI is not available, the browser’s built-in voice reads the same briefing.

4.9 Design System & UI Implementation

The website uses a dark “Eco-Obsidian” theme: a deep black-blue background with frosted-glass panels. Each energy source has its own colour, so colours carry meaning in cards, charts and 3D models. All styles are in one CSS file (about 44 KB). Hover effects, counting numbers and smooth page changes make the website feel lively.

Table 4.3: Colours used

Use	Colour	Code
Background	Deep carbon black	#070b12 / #0c121d
Panels	Frosted glass	rgba(11, 16, 26, 0.82)
Solar	Golden amber	#f59e0b
Wind	Cyan	#06b6d4
Hydro	Aquamarine	#0ea5e9 / #2dd4bf
Biomass	Emerald green	#10b981
Battery	Violet / lime	#8b5cf6 / #22c55e
Grid	Alert coral red	#ef4444

CHAPTER 5
PHYSICS FORMULAS & MATHEMATICAL MODELS

This chapter lists the formulas used in the project. Each formula is followed by an "In simple words" line. The website uses the simplified daily versions so that results appear instantly.

5.1 Solar PV Power Formula

$$P(t) = P_{STC} \times (G / 1000) \times [1 - 0.0038\,(T_{cell} - 25)] \times \eta_{inv} \quad (5.1)$$

Here P_{STC} is the panel rating, G is the sunlight strength (W/m²), T_{cell} is the panel temperature (°C) and η_{inv} is the inverter efficiency. **In simple words:** more sunlight gives more power, but a hot panel loses about 0.38 % of its power for every degree above 25 °C. The simplified daily version is:

$$E_{solar}\ (kWh/day) = P_{cap} \times H_{sun} \times (\eta_{panel} / 100) \quad (5.2)$$

5.2 Wind Turbine Power (Betz Limit)

$$P_{wind} = ½ \times \rho \times A \times v^3 \times C_p \quad (5.3)$$

Here ρ is air density (about 1.225 kg/m³), A is the area swept by the blades, v is wind speed and C_p is the share of wind energy that the turbine can capture (never more than 0.593, the Betz limit). **In simple words:** if the wind speed doubles, the power becomes eight times. The turbine behaves in four ways depending on wind speed:

Table 5.1: Turbine behaviour

Situation	Wind speed	What happens
Standby	< 3 m/s	Too little wind; no power
Generating	3–12 m/s	Power rises quickly with wind
Full power	12–25 m/s	Power stays at the rated value
Storm brake	> 25 m/s	Brakes stop the turbine for safety

$$E_{wind}\ (kWh/day) = P_{turbine} \times N \times 24 \times (CF / 100) \quad (5.4)$$

5.3 Hydropower Formula

$$P_{hydro} = \eta_{total} \times \rho_w \times g \times Q \times H \quad (5.5)$$

Here ρ_w = 1000 kg/m³ (water), g = 9.81 m/s², Q is the water flow (m³/s), H is the height of fall (m) and η_{total} is the plant efficiency. **In simple words:** more water and a higher fall give more power. The daily version used is:

$$E_{hydro}\ (kWh/day) = P_{cap} \times t_{op} \times (\eta / 100) \quad (5.6)$$

5.4 Biomass LHV & CHP Model

$$LHV_{net} = LHV_{dry} \times (1 - M) - 0.68 \times M \quad (5.7)$$

$$E_{bio}\ (kWh/day) = m \times 4.5 \times (\eta_{conv} / 100) \quad (5.8)$$

Here M is the moisture share, m is the daily waste in kg, and 4.5 kWh/kg is the energy in dry biomass. **In simple words:** wet fuel gives less energy because part of the heat is used to dry the water out. Using the waste heat as well (CHP) adds about 46 %, so the total efficiency goes above 72 %.

5.5 Battery Energy Balance

Let R be the clean energy produced and D the demand for the day.

- If R is more than D: the extra (R − D) charges the battery, with a small loss. Anything beyond a full battery is exported or wasted.
- If R is less than D: the battery supplies the shortage first, with a small loss. If still short, the grid supplies the rest.

$$E_{grid} = \max(0, D - R - E_{batt,out}) \quad (5.9)$$

5.6 CO_2 Calculation Model (Indian Grid Factors)

Table 5.2: CO_2 released per unit of electricity

Source	kg CO_2 per kWh
Grid (India)	0.82
Solar PV	0.02
Wind	0.01
Hydro	0.04
Biomass	0.23

$$CO_{2avoided} = (D \times 0.82) - \Sigma (E_i \times f_i) \quad (5.10)$$

In simple words: CO_2 avoided is what the grid would have released for our demand, minus what our clean sources released. E_i is energy from source i and f_i is its CO_2 factor.

5.7 Cost Savings Model

$$Saving = (D \times T) - (E_{grid} \times T) \quad (5.11)$$

$$Saving\ \% = [Saving / (D \times T)] \times 100 \quad (5.12)$$

T is the grid price (₹/kWh). **In simple words:** saving is the bill we would pay with only the grid, minus the bill we still pay after using clean energy.

5.8 SDG 7 Scoring Algorithm

The SDG 7 score is out of 100 and is made of four parts, each worth 25 marks:

Table 5.3: SDG 7 score parts

Part	Maximum	Depends on
Renewable Adoption	25	Share of renewable energy (%)
Clean Energy	25	CO_2 avoided
Energy Efficiency	25	How little the grid is needed
Affordability	25	Money saved
Total	**100**	

CHAPTER 6
RESULTS & DISCUSSION

6.1 System Performance Overview

GreenPulse updates every result as soon as a slider is moved, with no waiting, because each calculation is very short. All pages read the same data, so the Dashboard, Simulator, diagram, 3D model and voice briefing always agree. Below is a worked example using the default settings of Table 4.1 and the formulas of Chapter 5.

6.2 Sample Simulation Results (Default Parameters)

Table 6.1: Daily energy with default settings

Source	Calculation	Energy (kWh/day)
Solar PV	10 × 5.5 × 0.18	9.90
Wind	5 × 2 × 24 × 0.30	72.00
Hydro	3 × 8 × 0.80	19.20
Biomass	50 × 4.5 × 0.25	56.25
Total clean energy		**157.35**

The daily demand is 60 kWh, and the system makes about 2.6 times that. So the whole demand is met by clean energy, the grid is not needed (0 %), and the saving is the full bill: 60 × ₹7 = ₹420 per day. The extra 97.35 kWh first fills the 20 kWh battery (it starts half full), and the rest can be exported.

Table 6.2: Other results for the default settings

Result	Value
Clean energy share of demand	100 %
Grid import	0 kWh/day
Bill with grid only (60 kWh × ₹7)	₹420/day
Money saved	₹420/day (100 %)
CO_2 if grid only (60 × 0.82)	49.20 kg/day
CO_2 from our clean mix (0.093 kg/kWh × 60 kWh)	5.58 kg/day
CO_2 avoided	43.62 kg/day

Note: this is an example calculation made by hand. The numbers shown live in the website may differ a little because of how extra energy is counted.

6.3 SDG 7 Impact Measurement

The SDG 7 meter turns all results into one number. With the default settings, all demand is met by clean energy, the full bill is saved and a lot of CO_2 is avoided, so the score is very high (93/100 was seen on the dashboard during development). If the user reduces the generation, increases the demand or uses more biomass (which gives more CO_2 than solar or wind), the score falls. This helps students see which decisions matter most for clean energy goals.

6.4 Energy Audit Case Study

To show the audit page, we take a small example facility with a price of ₹7/kWh and the equipment in Table 6.3 (example values).

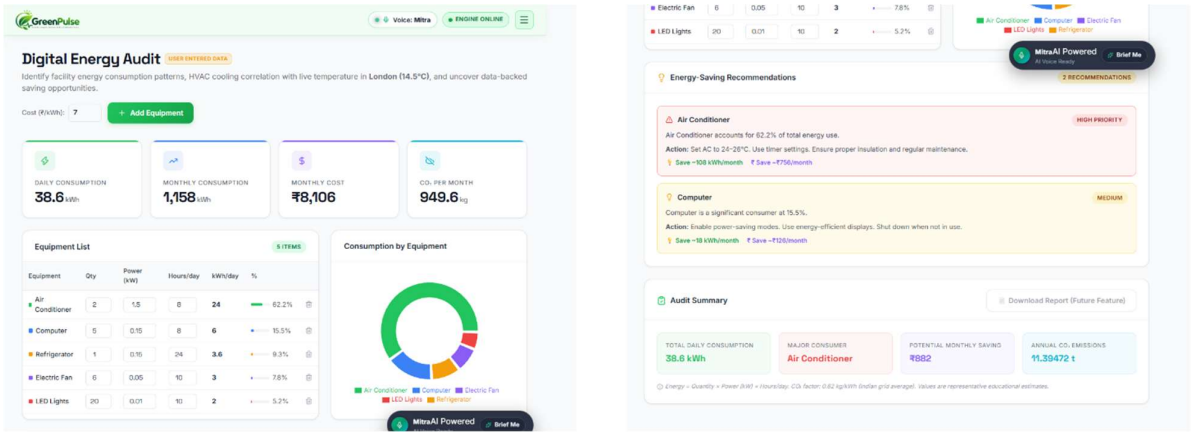


Figure 6.1: Energy Audit Page

Table 6.3: Example energy audit

Equipment	Qty	kW each	Hours/day	kWh/day	Share
Air conditioner	3	1.50	6	27.0	62.2 %
Computer	5	0.20	8	8.0	18.4 %
Ceiling fan	10	0.07	8	5.6	12.9 %
LED light	20	0.02	7	2.8	6.5 %
Total				**43.4**	**100 %**

The facility uses 43.4 kWh a day, or 1302 kWh a month. At ₹7 per unit this costs ₹9,114 a month and releases about 1068 kg of CO_2 (1302 × 0.82). The air conditioners use 62.2 % of the energy, so the page marks them as the first thing to fix. If the AC thermostat is set a little higher and the units are serviced, about 10 % of their use can be saved, which is 81 kWh or ₹567 every month.

6.5 Technology Comparison Insights

The comparison page shows that no technology is best in everything. Wind, hydro and nuclear release the least CO_2. Solar is cheapest to set up among clean sources but works only in daytime. Hydro is available most of the time. Coal and gas are cheap to build but release about 820 and 490 g CO_2 per kWh. Biomass can run on demand but releases more CO_2 than other renewables. Green hydrogen is clean but loses a lot of energy during conversion.

Table 6.4: Approximate values used for comparison (for learning)

Technology	g CO_2/kWh	Capacity factor	Set-up cost (₹/kW)
Solar PV	~40	15–25 %	40,000–50,000
Wind	~11	25–40 %	60,000–70,000
Hydropower	~24	40–60 %	1,00,000–1,50,000
Biomass	~230	60–80 %	60,000–90,000
Coal	~820	60–85 %	70,000–90,000
Nuclear	~12	80–90 %	1,50,000–2,00,000
Natural gas	~490	40–60 %	45,000–55,000
Green hydrogen	Low (depends on power used)	Depends on supply	> 1,50,000

CO_2 values follow IPCC lifecycle estimates. The other values are approximate ranges for Indian conditions, and the website's data may differ slightly.

6.6 3D Digital Twin Demonstration

In the 3D model, adding more wind turbines instantly adds more turbines to the scene. Night mode darkens the campus so the battery LEDs stand out, and the camera buttons let the user visit each part. The 3D model turns slider numbers into a real-looking layout, which helps students understand size and space.

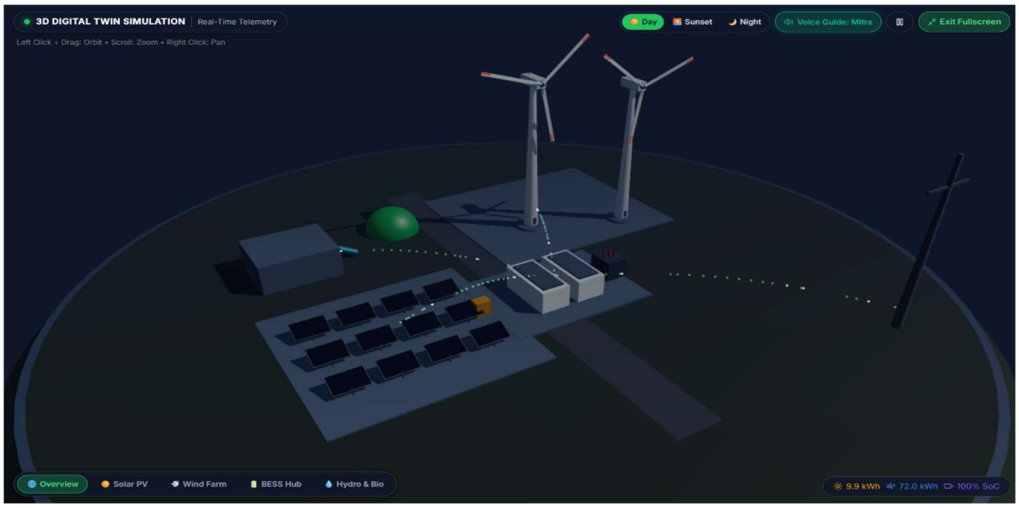
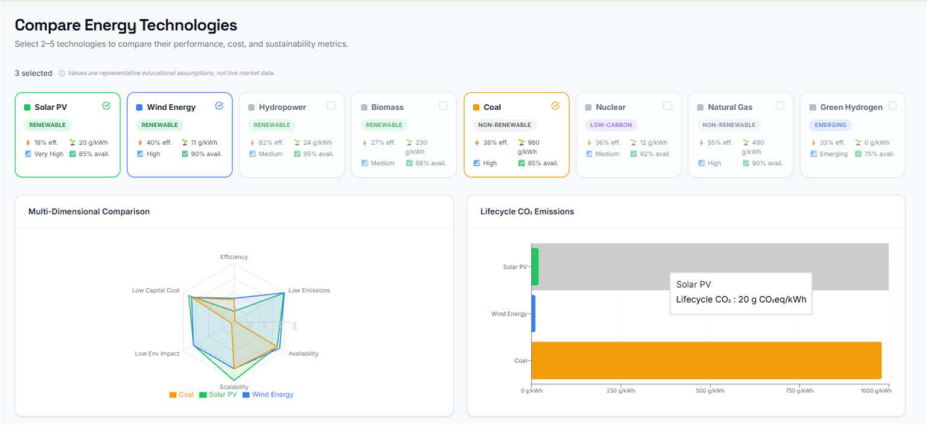


Figure 6.2: 3D Digital Twin in Daylight, Sunset and Night modes



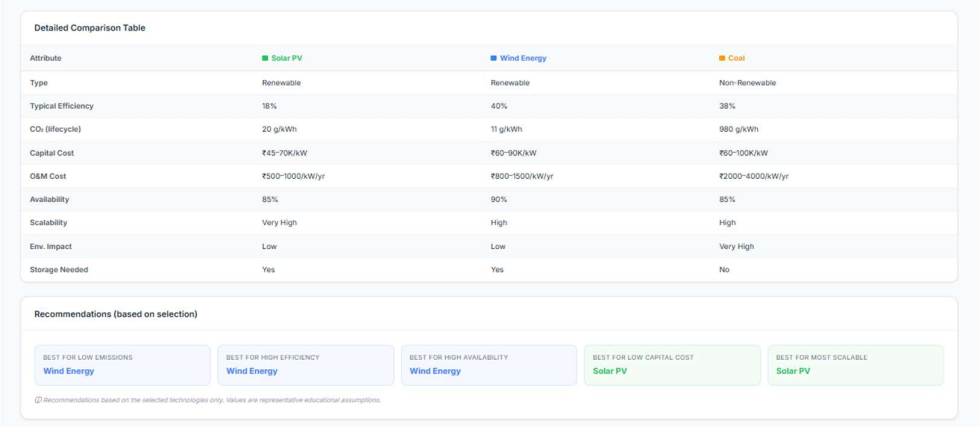


Figure 6.3: Comparison of energy Technology

CHAPTER 7
TESTING

7.1 Testing Approach

We tested the project by hand. We used every page, tried normal, smallest and largest values, and compared the results with our own calculations from Chapter 5. We also ran OxLint to find code mistakes. Automatic tests (Vitest) have not been added yet and are part of the future work. The *Result* column in the tables below is to be filled in by the team after the final test.

7.2 Functional Testing

Table 7.1: Feature test cases

ID	Feature	What we do	What should happen
F1	Dashboard	Change solar capacity slider	Cards, chart and SDG 7 score update at once
F2	Simulator	Set all generation to 0	Generation is 0; grid dependency 100 %
F3	Simulator	Set demand below generation	Surplus shown; battery charges; no grid import
F4	Battery	Set capacity to 0 kWh	No charging or discharging; no error
F5	Digital Twin	Click each camera view	Camera moves to that area
F6	Digital Twin	Switch lighting modes	Day, sunset and night look correct
F7	Hydrogen	Enter 100 kWh/day at 70 %	Hydrogen output and round-trip efficiency shown
F8	Waste-to-Energy	Change the process	Results and chemistry update
F9	Energy Audit	Add, edit and delete equipment	Totals, bill and chart update
F10	Compare	Select 2 and 5 technologies	Table and badges update; limits respected
F11	Voice	Press Brief Me with and without API key	Gemini voice, or browser voice as backup

7.3 UI/UX Testing

We checked that text is easy to read on the dark panels, that colours mean the same thing everywhere, that the layout adjusts to different window sizes, that page changes are smooth and that every slider has a clear name and unit. The design is best on laptops and large tablets. A mobile-first design is planned for the future.

7.4 Cross-browser Compatibility

GreenPulse uses standard browser features: WebGL for 3D, full screen mode and the built-in speech voice. These work in current versions of the main browsers.

Table 7.2: Browser checklist

Browser	3D model	Full screen	Backup voice
Google Chrome	Supported	Supported	Supported
Microsoft Edge	Supported	Supported	Supported
Mozilla Firefox	Supported	Supported	Supported (voice varies)
Apple Safari	Supported	Supported	Supported (voice varies)

7.5 Performance Testing

The website is fast because calculations are tiny, results are reused when nothing changes, the build is optimised by Vite and Vercel serves the files from nearby servers. The 3D scene is the heaviest part, and its smoothness depends on the computer's graphics. We suggest noting the page load time (from a Lighthouse test) and the 3D frame rate on a normal laptop, and adding them here.

CHAPTER 8
CONCLUSION & FUTURE SCOPE

8.1 Summary of Work Done

In this project we built and published GreenPulse, an eight-page website for learning about the clean energy transition. Ten calculation files cover solar, wind, hydro, biomass, battery, hydrogen, waste-to-energy, energy audit, SDG score and voice. A 3D model, an animated diagram and charts show the results, and the Mitra and Miles voice assistant explains them. The whole website runs in the browser, is open-source (MIT licence) and is live on Vercel.

8.2 Objectives Achieved

Table 8.1: Objectives and how they were met

Objective	How it was achieved
Clean energy + battery simulator	Simulator page with all five parts
Simple separate calculations	Ten small calculation files
Visual explanation	3D Digital Twin and animated flow diagram
Measuring sustainability	CO_2, savings and SDG 7 score on Dashboard
Syllabus coverage	One module for each of Units 1 to 5
Technology comparison	Eight-technology table with badges
Voice help and publishing	Mitra & Miles voice assistant; Vercel hosting

8.3 Limitations

- The calculations are simplified for learning and are not exact engineering results.
- Settings are lost when the page is refreshed.
- Only one day is simulated; seasons and years are not covered.
- The wind speed slider is only for show and does not change the wind power yet.
- PDF/CSV download of results is not available yet (the button is disabled).
- The code uses plain JavaScript (no TypeScript) and has no automatic tests.

8.4 Future Enhancements

1. Connect wind speed to the real power curve (power grows with speed cubed).
2. Simulate many days and seasons using weather data.
3. Add PDF/CSV report download.
4. Add a cost calculator (LCOE, NPV, IRR).
5. Save settings in the browser (localStorage).
6. Move the code to TypeScript.
7. Add automatic tests with Vitest.
8. Make the design mobile-first.
9. Save and compare three designs side by side.
10. Use live data from public energy websites.

REFERENCES

A. GTU Syllabus

[1] Gujarat Technological University, *Syllabus: Sustainable Energy Technology (BE05000561)*, https://www.gtu.ac.in.

[2]

B. Languages & Technologies Used

[1] JavaScript (ES2022+) and JSX — MDN Web Docs, https://developer.mozilla.org/en-US/docs/Web/JavaScript

[2] React 19 — https://react.dev.

[3] React Router v7 — https://reactrouter.com.

[4] Vite 8 (build tool) — https://vite.dev.

[5] Three.js (WebGL 3D graphics) — https://threejs.org/docs.

[6] GSAP and @gsap/react (animation) — https://gsap.com/docs.

[7] Framer Motion / Motion (page transitions) — https://motion.dev.

[8] Lenis (smooth scrolling) — https://lenis.darkroom.engineering.

[9] Recharts (charts) — https://recharts.org.

[10] Lucide React (icons) — https://lucide.dev.

[11] react-countup (animated numbers) — https://github.com/glennreyes/react-countup.

[12] TailwindCSS v4 and CSS — https://tailwindcss.com; https://developer.mozilla.org/en-US/docs/Web/CSS.

[13] OxLint (code linter) — https://oxc.rs.

[14] Google Gemini API (AI voice) — https://ai.google.dev.

[15] Web Speech API (browser voice) — https://developer.mozilla.org/en-US/docs/Web/API/Web_Speech_API.

[16] Vercel (hosting and deployment) — https://vercel.com/docs.

[17] GitHub (source code): https://github.com/Vinay-vaja/SET_project. Live project: https://set-visulizer-mopec-ldce.vercel.app.

APPENDIX

A. Project Directory Structure

```
SET_project/
├── public/              favicon.ico, favicon.png, favicon.svg,
│                        apple-touch-icon.png, logo.png, logo-icon.png,
│                        hero-landscape.jpg, icons.svg
├── Project Report/
│   └── Greenpulse-Sustainable_Enery_Transition.pdf
├── src/
│   ├── main.jsx, App.jsx, App.css, index.css
│   ├── context/         EnergyContext.jsx, VoiceContext.jsx,
│   │                    RealDataContext.jsx
│   ├── data/            energyConstants.js, sdgData.js, technologies.js
│   ├── hooks/           useRealConditions.js
│   ├── services/        locationService.js, weatherService.js,
│   │                    solarDataService.js, realDataService.js
│   ├── utils/           solarCalculations.js, windCalculations.js,
│   │                    hydroCalculations.js, biomassCalculations.js,
│   │                    batteryCalculations.js, sdgScore.js,
│   │                    auditCalculations.js, hydrogenCalculations.js,
│   │                    wasteCalculations.js, geminiVoiceService.js
│   ├── components/      Navbar.jsx, KPICard.jsx, AnimatedNumber.jsx,
│   │                    SliderInput.jsx, EnergyFlowDiagram.jsx,
│   │                    DigitalTwin3D.jsx, GenerationProcessViewer.jsx,
│   │                    InteractiveEcoSphere.jsx, SDGScoreRing.jsx,
│   │                    VoiceAssistantWidget.jsx
│   │   └── simulation/  SimulationViewport.jsx, SolarSimulationView.jsx,
│   │                    WindSimulationView.jsx, HydroSimulationView.jsx,
│   │                    BiomassSimulationView.jsx
│   └── pages/           Dashboard.jsx, Simulator.jsx, EnergyAudit.jsx,
│                        Compare.jsx, SDGImpact.jsx, EmergingTech.jsx,
│                        WasteToEnergy.jsx, About.jsx
├── index.html, vite.config.js, package.json
└── .oxlintrc.json, PROJECT_ANALYSIS.txt, .gitignore
```

B. Setup & Installation Guide

1. Install Node.js (the current LTS version, which includes npm) and Git.
2. Download the code: git clone https://github.com/Vinay-vaja/SET_project.git and open the folder.
3. Install the libraries with npm install.
4. (Optional) Add a Google Gemini API key as explained in the repository README for the cloud voice. Without it, the browser voice is used.
5. Start the project with npm run dev and open the address shown in the browser.
6. To make the final version, run npm run build, and to view it, run npm run preview.

C. Abbreviations & Glossary

Table C.1: Abbreviations

Term	Meaning	Term	Meaning
AFC	Alkaline fuel cell	PAFC	Phosphoric acid fuel cell
AWE	Alkaline water electrolyzer	PEM	Proton exchange membrane
BESS	Battery energy storage system	PV	Photovoltaic (solar)
CHP	Combined heat and power	SDG	Sustainable Development Goal
EMS	Energy management system	SET	Sustainable Energy Technology
GTU	Gujarat Technological University	SMR	Steam methane reforming
LCOE	Levelised cost of energy	SOFC	Solid oxide fuel cell
LHV	Lower heating value	SPA	Single-page application
SOEC	Solid oxide electrolyzer cell	STC	Standard test conditions
SoC	State of charge	WtE	Waste-to-energy